

Playwright Automation Framework – Test Report

1. Project Title

UI Automation Testing Framework using Playwright

2. Objective

The objective of this project is to design and implement a scalable UI automation framework using Playwright and TypeScript.

Target Website:

<https://www.retailmart.linkpc.net>

3. User Story

User Story 1 – Search + Price Filter Validation

As a user,

I want to search for a product and apply price filter

User Story 2 – Payment Processing Error Validation

Verify system behavior during payment processing error. Acceptance Criteria:

4. Environment & Setup

Hardware

- System: Windows
- RAM: Minimum 8 GB
- Processor: Intel i5 or above

Software

Component Version

Node.js Latest LTS

Playwright Latest

TypeScript Latest

Component Version

VS Code Latest

Browser Chromium

5. Framework Design

Framework Architecture:

Fixture Layer

↓

Flow Layer

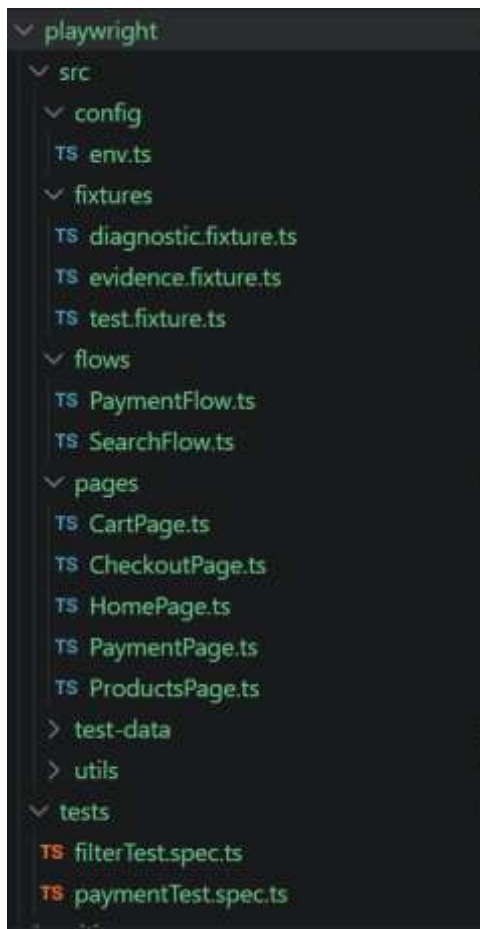
↓

Page Layer

↓

Test Layer

6. Project Structure



7. Scenarios Covered

Scenario 1 – Search for "laptop" and apply a price filter.

Input:

Search → “laptop”

Status:

PASS

Scenario 2 – Verify system behavior during payment processing error.

Status:

PASS

8. Execution Steps

Install dependencies:

npm install

Install browsers:

npx playwright install

Execute tests:

npx playwright test website-tests2/tests --headed

Open report:

npx playwright show-report

9. Test Results

Execution Summary:

Total Tests: 2

Passed: 2

Failed: 0

Execution Mode:

Chromium Browser

Execution Duration:
~14 seconds

10. Evidence

The screenshot displays the Cypress test runner interface. At the top, the test title is "Searching for product and applying filter" with a duration of 6.6s. Below the title, the file path "filterTest.spec.ts:7" is shown. A "chromium" button and a "Run" button are visible. The "Test Steps" section is expanded, showing a list of 20 steps with their respective durations. The steps include navigation, assertions, and interactions with the application.

Step	Duration
Before Hooks	2.8s
Navigate to "/" — ./src/pages/HomePage.ts:17	2.8s
Expect "toBeVisible" getByTestId('search-input') — ./src/pages/HomePage.ts:22	73ms
Expect "toBeVisible" getByTestId('hero-banner') — ./src/pages/HomePage.ts:23	27ms
Expect "toBeVisible" getByTestId('search-button') — ./src/pages/HomePage.ts:24	21ms
Fill "laptop" getByTestId('search-input') — ./src/pages/HomePage.ts:28	73ms
Click getByTestId('search-button') — ./src/pages/HomePage.ts:29	207ms
Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23	25ms
Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:27	224ms
Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30	13ms
Click getByTestId('price-filter-4') — ./src/pages/ProductsPage.ts:34	120ms
Evaluate locator('[data-testid^=\product-price-\T'] — ./src/pages/ProductsPage.ts:38	61ms
Expect "toBeGreaterThanOrEqual" — ./src/pages/ProductsPage.ts:42	0ms
Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23	25ms
Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:25	20ms
Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30	14ms
After Hooks	1.2s

Payment error validation

paymentTest.spec.ts:5

6.7s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks	3.2s
> ✓ Navigate to "/" — ./src/pages/HomePage.ts:17	4.2s
> ✓ Expect "toBeVisible" getByTestId('search-input') — ./src/pages/HomePage.ts:22	109ms
> ✓ Expect "toBeVisible" getByTestId('hero-banner') — ./src/pages/HomePage.ts:23	25ms
> ✓ Expect "toBeVisible" getByTestId('search-button') — ./src/pages/HomePage.ts:24	27ms
> ✓ Fill "Laptop" getByTestId('search-input') — ./src/pages/HomePage.ts:26	78ms
> ✓ Click getByTestId('search-button') — ./src/pages/HomePage.ts:29	173ms
> ✓ Expect "toHaveURL" — ./src/pages/ProductsPage.ts:23	26ms
> ✓ Expect "toContainText" getByTestId('search-results-heading') — ./src/pages/ProductsPage.ts:27	30ms
> ✓ Expect "toBeVisible" getByText('Category') — ./src/pages/ProductsPage.ts:30	51ms
> ✓ Click locator('[data-testid^="add-to-cart-"]')first() — ./src/pages/ProductsPage.ts:50	115ms
> ✓ Click getByTestId('cart-count') — ./src/pages/ProductsPage.ts:55	136ms
> ✓ Expect "toBeVisible" getByRole('heading', { name: 'Shopping Cart' }) — ./src/pages/CartPage.ts:15	116ms
> ✓ Expect "toBeVisible" getByTestId('checkout-button') — ./src/pages/CartPage.ts:16	21ms
> ✓ Click getByTestId('checkout-button') — ./src/pages/CartPage.ts:20	118ms
> ✓ Expect "toBeVisible" getByRole('heading', { name: 'Checkout', exact: true }) — ./src/pages/CheckoutPage.ts:29	71ms
> ✓ Expect "toBeVisible" getByTestId('continue-to-payment-button') — ./src/pages/CheckoutPage.ts:30	20ms
> ✓ Fill "Mani" getByTestId('guest-name-input') — ./src/pages/CheckoutPage.ts:34	61ms
> ✓ Fill "mani@gmail.com" getByTestId('guest-email-input') — ./src/pages/CheckoutPage.ts:35	51ms
> ✓ Fill "9876543210" getByTestId('guest-phone-input') — ./src/pages/CheckoutPage.ts:36	64ms
> ✓ Fill "Kulathoor" getByTestId('shipping-street-input') — ./src/pages/CheckoutPage.ts:37	62ms
> ✓ Fill "Trivandrum" getByTestId('shipping-city-input') — ./src/pages/CheckoutPage.ts:38	58ms
> ✓ Select option getByTestId('shipping-state-select') — ./src/pages/CheckoutPage.ts:39	45ms
> ✓ Fill "62701" getByTestId('shipping-zip-input') — ./src/pages/CheckoutPage.ts:40	48ms
> ✓ Click getByTestId('continue-to-payment-button') — ./src/pages/CheckoutPage.ts:44	104ms
> ✓ Expect "toBeVisible" getByRole('heading', { name: 'Payment', exact: true }) — ./src/pages/PaymentPage.ts:25	52ms
> ✓ Expect "toBeVisible" getByTestId('place-order-button') — ./src/pages/PaymentPage.ts:26	9ms
> ✓ Fill "Mani" getByTestId('payment-card-name') — ./src/pages/PaymentPage.ts:30	48ms
> ✓ Fill "4111 1111 1111 1111" getByTestId('payment-card-number') — ./src/pages/PaymentPage.ts:31	36ms
> ✓ Fill "12/28" getByTestId('payment-expiry') — ./src/pages/PaymentPage.ts:32	47ms
> ✓ Fill "456" getByTestId('payment-cvv') — ./src/pages/PaymentPage.ts:33	41ms
> ✓ Click getByTestId('place-order-button') — ./src/pages/PaymentPage.ts:37	52ms
> ✓ Expect "toHaveText" getByTestId('payment-general-error') — ./src/pages/PaymentPage.ts:42	12ms
> ✓ After Hooks	664ms

11. Environment Configuration

Playwright Configuration:

- **Browser: Chromium**
- **Trace Collection: Enabled**
- **Base URL Configured**
- **HTML Reporting Enabled**

12. Conclusion

The Playwright automation framework was successfully implemented using Page Object Model, Flow Layer, and Fixtures.